

Gian Marco Franceschini

POSTDOCTORAL RESEARCHER · COMPUTATIONAL BIOLOGY

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EXPERIENCE	Postdoctoral Researcher	NOV 2023 – PRESENT
	University of Lausanne — Lab. of Computational System Oncology, DBC · PI: Prof. Giovanni Ciriello · Lausanne, Switzerland	
	<i>Method development and analysis of chromatin compartments in cancer evolution through Hi-C and single-cell Hi-C.</i>	
	Postdoctoral Researcher	JAN – OCT 2023
	University of Trento — Lab. of Computational and Functional Oncology, CIBIO · PI: Prof. Francesca Demichelis · Trento, Italy	
	<i>Detecting neuroendocrine prostate cancer through cell-free DNA methylation profiling.</i>	
	PhD Candidate in Biomolecular Sciences	NOV 2018 – JAN 2023
	University of Trento — Lab. of Computational and Functional Oncology, CIBIO · PI: Prof. Francesca Demichelis · Trento, Italy	
	<i>Thesis: The DNA methylation landscape of metastatic prostate cancer: from characterization to liquid biopsy applications.</i>	
	Visiting Researcher — EMBO Exchange Grant	NOV 2021 – FEB 2022
EDUCATION	University of Lausanne — Lab. of Computational System Oncology · PI: Prof. Giovanni Ciriello · Lausanne, Switzerland	
	<i>Chromatin compartments in colorectal cancer cell lines under pharmacological or genetic perturbation of the DNA methylation machinery.</i>	
	Student Assistant — RNA-seq Data Analysis	SEPT 2017 – SEPT 2018
	Lab. of Experimental Cancer Biology, CIBIO · PI: Prof. Maria Caterina Mione · Trento, Italy	
	<i>RNA-seq data analysis from zebrafish disease models of cancer.</i>	
	Tutor — Bachelor Courses in Biomolecular Sciences	NOV 2017 – JUL 2018
	CIBIO, University of Trento · Trento, Italy	
	<i>Computer Science (40 hrs): Unix, command line, Python. Computational Biology (20 hrs): sequencing analysis and biological data exploration.</i>	
	PhD cum Laude — Biomolecular Sciences, Computational Biology track	NOV 2018 – JAN 2023
	CIBIO, University of Trento · Trento, Italy	
PUBLICATIONS	MSc cum Laude — Quantitative and Computational Biology · 110/110	SEPT 2016 – OCT 2018
	CIBIO, University of Trento · Trento, Italy	
	<i>Internship: Mutual exclusivity and synthetic lethality in cancer DNA methylation patterns. PI: Prof. Francesca Demichelis.</i>	
	BSc cum Laude — Biomolecular Science and Technologies · 110/110	SEPT 2013 – JUL 2016
	CIBIO, University of Trento · Trento, Italy · GPA 29.5/30	
	<i>Internship: Cas9 delivery through HIV-1 derived Viral Like Particles. PI: Prof. Anna Cereseto.</i>	
	Liceo Scientifico · 100/100	2008 – 2013
	ISSIS G. Verdi · Valdobbiadene, Italy	
	NAT. REV. CANCER · 2026	
	Origins and consequences of oncogenic 3D chromatin remodeling	
	GM Franceschini* , K Georgakopoulou*, G Ciriello, E Oricchio	
	NATURE METHODS · 2026	
	TADShop: systematic benchmarking and identification of topologically associating domains	
	P Li*, A Hatos*, M Petrovic, GM Franceschini , L Nanni, D Tavernari, G Ciriello	
	NATURE COMMUN. · 2026	
	HaploC-tools reveal haplotype-specific chromosome conformation and chromatin states	
	Y Liu, GM Franceschini , Y Zhang, E Oricchio, G Ciriello	
	GENOME MEDICINE · 2026	
	A computational framework for sensitive tumor detection and accurate subtyping using shallow cell-free DNA methylome sequencing	
	M Paoli, F Galardi, A Nardone, C Biagioni, D Romagnoli, S Di Donato, GM Franceschini , L Livraghi, M Pestrin, G Sanna, E Risi, I Migliaccio, E Moretti, L Malorni, L Biganzoli, F Demichelis, M Benelli	
	CANCER DISCOVERY · 2024	
	Non-invasive detection of neuroendocrine prostate cancer through targeted cell-free DNA methylation	
	GM Franceschini , O Quaini, K Mizuno, F Orlando, Y Ciani, SY Ku, M Sigouros, E Rothmann, A Alonso, M Benelli, C Nardella, J Auh, D Freeman, B Hanratty, M Adil, O Elemento, ST Tagawa, F Feng, O Caffo, C Buttiglierio, U Basso, P Nelson, E Corey, MC Haffner, A Aparicio, F Demichelis, H Beltran	
	BRIEF. BIOINFORM. · 2023	
	MIMESIS: minimal DNA-methylation signatures to quantify and classify tumor signals in tissue and cell-free DNA samples	
	D Romagnoli, A Nardone, F Galardi, M Paoli, F De Luca, C Biagioni, GM Franceschini , M Pestrin, G Sanna, E Moretti, F Demichelis, I Migliaccio, L Biganzoli, L Malorni, M Benelli	
	COMMUN. BIOLOGY · 2021	
	Cancer methylomes characterization enabled by Rocker-meth	
	M Benelli, GM Franceschini* , A Magi, D Romagnoli, C Biagioni, I Migliaccio, L Malorni, A Di Leo, F Demichelis	
	CELL SYSTEMS · 2021	
	Allele-specific genomic data elucidate the role of somatic gain and copy-number neutral loss of heterozygosity in cancer	
	Y Ciani, T Fedrizzi, D Prandi, F Lorenzin, A Locallo, P Gasperini, GM Franceschini , M Benelli, O Elemento, LL Fava, A Inga, F Demichelis	
	J. CLIN. INVEST. · 2020	
	Circulating tumor DNA profile recognizes transformation to castration-resistant neuroendocrine prostate cancer	
	H Beltran, A Romanel, V Conteduca, N Casiraghi, M Sigouros, GM Franceschini , F Orlando, T Fedrizzi, SY Ku, E Dann, A Alonso, JM Mosquera, A Sboner, J Xiang, O Elemento, DM Nanus, ST Tagawa, M Benelli, F Demichelis	

Genome-wide plasma DNA methylation features of metastatic prostate cancer

A Wu, P Cremaschi, D Wetterskog, V Conteduca, **GM Franceschini**, D Klefogiannis, A Jayaram, S Sandhu, SQ Wong, M Benelli, S Salvi, G Gurioli, A Feber, M Buongiorno Pereira, A Wingate, E Gonzalez-Billalabeitia, U De Giorgi, F Demichelis, S Lise, G Attard

VSV-G enveloped vesicles for traceless delivery of CRISPR-Cas9

C Montagna, G Petris, A Casini, G Maule, **GM Franceschini**, I Zanella, L Conti, F Arnoldi, OR Burrone, L Zentilin, S Zacchigna, M Giacca, A Cereseto

GRANTS & FELLOWSHIPS	Novartis Freenovation Grant • CHF 198,000 <i>Multiscale chromatin blueprints to decipher and forecast epigenetic perturbations for cancer therapy</i> EMBO Scientific Exchange Grant #9295 • CHF 11,500 Research visiting, University of Lausanne Erasmus Scholarship Research visiting, University of Lausanne CSAMA 2018 — Bioinformatics Summer School Selected among 5 international students for a full scholarship Travel Grant — Vienna Biocenter Continuum Symposium Selected among 11 students from across Europe Scholarship — Club Alpbach Trentino European Forum Alpbach	NOV 2025 JAN 2022 NOV 2021 JUL 2018 NOV 2017 SUMMER 2017
AWARDS	S.O.S Onlus — Honorable Mention For scientific results in translational oncology obtained during PhD Fondazione 7 Ville — Scholarship Award For excellent results during Master of Science degree Merit Award — University of Trento For completing Bachelor's degree with a high GPA on the first available session Fondazione 7 Ville — Scholarship Award For excellent results during Bachelor of Science degree Robert Gallo e Renzo Girola ONLUS — Scholarship Protti Foundation — Scholarship For the outstanding high school path	NOV 2024 SEPT 2019 FALL 2017 SEPT 2017 2013 – 2018 2013
PATENT	Detection of neuroendocrine prostate cancer through cell-free DNA methylation sequencing Co-inventor • International Publication Number: WO 2021/162981 A2	2021
CONFERENCES	EMBO Cancer Evolution, Rimini Selected talk	2026
PEER REVIEW	Nature* Nature Genetics* (2) Nature Commun.* (1) *: assisting senior reviewer	
PROGRAMMING	R Advanced — package development, Bioconductor ecosystem, machine learning, statistical learning Python pandas, numpy, scipy, BioPython, scikit-learn, TensorFlow, Keras, PyTorch Bash Pipeline management and scripting Design Adobe suite, Affinity Designer, Inkscape, Procreate Other Git, $\text{\LaTeX}$, C# (Unity), Octave/Matlab, C — Ubuntu, Windows	
LANGUAGES	Italian — mother tongue English — C1 French — B1	
OUTREACH & LEADERSHIP	Telegram Educational Contest — Award Winner Placed 7th among over 7,000 entries with an educational quiz on cancer for mobile platforms Science Communication — <i>La medicina in uno scatto</i> Italian website with more than 100k followers promoting medical and scientific awareness Seminar Assistant — Precision Medicine and Its Impact on Our Society EFA 2020, Alpbach, Austria <i>Drafted and proposed a one-week seminar introducing future policy makers to precision medicine.</i> Board Member & IT Manager — Club Alpbach Trentino Fundraising for scholarships, event planning, and team coordination Student Representative — Center of Integrative Biology, UniTN Volunteering — VolontariInStrada Weekly gatherings assisting homeless individuals	AUG 2020 MAR 2016 – SEPT 2017 AUG 2018 AUG 2017 – NOV 2018 NOV 2016 – NOV 2018 FEB 2019 – JAN 2020